Lecture 1: Asymptotic Analysis

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1 Definition

$$
\begin{aligned}
&O(g) : \{f : \exists c > 0 \wedge n_0 > 0, \forall n \geq n_0, 0 \leq f(n) \leq c \cdot g(n)\} \\
&\Omega(g) : \{f : \exists c > 0 \wedge n_0 > 0, \forall n \geq n_0, 0 \leq c \cdot g(n) \leq f(n)\} \\
&\Theta(g) : \{f : \exists c_1, c_2 > 0 \wedge n_0 > 0, \forall n \geq n_0, c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)\} \\
&o(g) : \{f : \forall c > 0, \exists n_0 > 0, \forall n \geq n_0, 0 \leq f(n) < c \cdot g(n)\} \\
&\omega(g) : \{f : \forall c > 0, \forall n \leq n_0, 0 \leq c \cdot g(n) < f(n)\}
\end{aligned}
$$

2 Proof with limit

3 Secondary conclusions

$$\Theta(f) = O(f) \cap \Omega(f)$$

Proof. 1. prove: $O(f) \cap \Omega(f) \subseteq \Theta(f)$
From definition:

$\Theta(f(n)) : \{g : \exists c_1, c_2 > 0, \exists n_0 \geq 0, \forall n \geq n_0, c_1 \cdot f(n) \leq g(n) \leq c_2 \cdot f(n)\}$
$O(f(n)) \cap \Omega(f(n)) : \{g : \exists c_1 > 0, \exists n_1 \geq 0, \forall n \geq n_1, 0 \leq g(n) \leq c_1 \cdot f(n) \wedge \exists c_2 > 0, \exists n_2 \geq 0, \forall n \geq n_2, 0 \leq c_2 \cdot f(n) \leq g(n)\}$

When $0 \leq c_1 \leq c_2, O(f(n)) \cap \Omega(f(n)) : \{g : \exists n_1, n_2 \geq 0, \forall n \geq \{\max(n_1, n_2)\}, 0 \leq c_1 f(n) \leq g(n) \leq c_2 f(n)\}$
$\implies O(f) \cap \Omega(f) \subseteq \Theta(f)$
2. prove: $\Theta(f) \subseteq O(f) \cap \Omega(f)$
$(f(n) > 0 \wedge, c_1 \leq, c_2) \implies (0 \leq c_1 f(n) \leq g(n) \wedge 0 \leq g(n) \leq c_2 f(n))$
$\implies \Theta(f) \subseteq O(f) \cap \Omega(f)$
$\implies \Theta(f) = O(f) \cap \Omega(f)$

□

$$\lim_{n \to \infty} \frac{f(n)}{g(n)} = 0 \implies f(n) \in o(g(n))$$

Proof. From lmit definition, $\forall \epsilon > 0, \exists n_0 \geq 0, \forall n \geq n_0, \frac{f(n)}{g(n)} < \epsilon$. Let $c = \epsilon, \frac{f(n)}{g(n)} < \epsilon, f(n) < c \cdot g(n)$.
$f(n) \geq 0, f(n) \in o(g(n))$ □

$$\lim_{n \to \infty} \frac{f(n)}{g(n)} = \infty \implies f(n) \in \omega(g(n))$$

Proof. From $\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty \iff \lim_{n\to\infty} \frac{g(n)}{f(n)} = 0$, as
$\lim_{x\to\infty} \frac{1}{f(x)} = \frac{1}{\lim_{x\to\infty} f(x)}, \lim_{n\to\infty} \frac{g(n)}{f(n)} = \frac{1}{\lim_{n\to\infty} \frac{f(n)}{g(n)}} = \frac{1}{\infty} = 0.$
Similarly, $\forall \epsilon > 0, \exists n_0 \geq 0, \forall n \geq n_0, \frac{g(n)}{f(n)} < \epsilon$. Let $c = \frac{1}{\epsilon}, \frac{g(n)}{f(n)} < \epsilon, c \cdot g(n) < f(n)$.

□

$$0 < \lim_{n\to\infty} \frac{f(n)}{g(n)} < \infty \implies f(n) \in \Theta(g(n))$$

Proof. Definitions:
$\Theta(g) : \{f : \exists c_1, c_2 > 0 \wedge n_0 > 0, \forall n \geq n_0, c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)\}$
$O(g) : \{f : \exists c > 0 \wedge n_0 > 0, \forall n \geq n_0, 0 \leq f(n) \leq c \cdot g(n)\}$
$o(g) : \{f : \forall c > 0, \exists n_0 > 0, \forall n \geq n_0, 0 \leq f(n) < c \cdot g(n)\}$
Prove: $\Theta(f) = O(f) \setminus o(f)$

Proof. 1. $\Theta(f) \subseteq O(f) \setminus o(f)$
Suppose $g(n) \in \Theta(f), \Theta(f) = O(f) \cap \Omega(f) \implies g(n) \in O(f)$.
Suppose $g(n) \in o(f), (\exists c_1, c_2 > 0, \exists n_1 > 0, \forall n \geq n_1, 0 \leq c_1 \cdot f(n) \leq g(n) \leq c_2 \cdot f(n)) \wedge (\forall c_3 > 0, \exists n_2 > 0, \forall n \geq n_2, 0 \leq g(n) < c_3 \cdot f(n))$
Let $c_3 < c_1, c_1 \cdot f(n) \leq g(n) < c_3 \cdot f(n)$, contradiction,
$\implies g(n) \in \Theta(f) \to g(n) \notin o(f)$
$\implies \Theta(f) \subseteq O(f) \setminus o(f)$
2.

$$O(f) \setminus o(f) \subseteq \Theta(f) =$$

$$\{g : (\exists c_1 > 0, \exists n_1 > 0, \forall n \geq n_1,\ 0 \leq g(n) \leq c_1 \cdot f(n)) \\ \wedge\ \neg(\forall c_2 > 0, \exists n_2 > 0, \forall n \geq n_2,\ 0 \leq g(n) < c_2 \cdot f(n))\}$$

$$\{g : (\exists c_1 > 0, \exists n_1 > 0, \forall n \geq n_1,\ 0 \leq g(n) \leq c_1 \cdot f(n)) \\ \wedge\ (\exists c_2 > 0, \forall n_2 > 0, \exists n \geq n_2,\ (g(n) < 0 \vee g(n) \geq c_2 \cdot f(n)))\}$$

$$\{g : \exists c_1 > c_2 > 0, \forall n_2 > 0, \exists n_1 \geq n_2, \forall n \geq n_1,\ 0 \leq c_2 \cdot f(n) \leq g(n) \leq c_1 \cdot f(n)\}$$

$$\{g : \exists c_1 > c_2 > 0, \exists n_0 > 0, \forall n \geq n_0,\ 0 \leq c_2 \cdot f(n) \leq g(n) \leq c_1 \cdot f(n)\}$$

□

□

4 Reflexivity

4.1 $f(n) \in O(f(n))$

Proof. Suppose $f(n) > 0$
Let $c = 1, n_0 = 1, \forall n \geq n_0, 0 \leq f(n) \leq f(n), \implies f(n) \in O(f(n))$ □

4.2 $f(n) \in \Omega(f(n))$, $f(n) \in \Theta(f(n))$

Similarly, by picking the right c and n_0 it's easy to prove existential statements.

5 Transitivity

5.1 $f(n) \in O(g(n)) \wedge g(n) \in O(h(n)) \implies f(n) \in O(h(n))$

Proof.

$$\exists c_1 > 0, n_1 > 0, \forall n \geq n_1, 0 \leq f(n) \leq c_1 \cdot g(n)$$
$$\exists c_2 > 0, n_2 > 0, \forall n \geq n_2, 0 \leq g(n) \leq c_2 \cdot h(n)$$

Let $c_1 = 1, n_0 = \max(n_1, n_2)$, $0 \leq f(n) \leq g(n) \leq c_2 \cdot h(n) \implies \exists c_2 > 0, \forall n \geq n_0, 0 \leq f(n) \leq c_2 \cdot h(n)$
$\implies f(n) \in O(h(n))$ □

5.2 $f(n) \in \Omega(g(n)) \wedge g(n) \in \Omega(h(n)) \implies f(n) \in \Omega(h(n))$

Similar to $O(f(n))$ proof

5.3 $f(n) \in \Theta(g(n)) \wedge g(n) \in \Theta(h(n)) \implies f(n) \in \Theta(h(n))$

$$\exists c_1^{[1]}, c_2^{[1]} > 0, n_1 > 0, \forall n \geq n_1, 0 \leq c_1^{[1]} \cdot g(n) \leq f(n) \leq c_2^{[1]} \cdot g(n)$$
$$\exists c_1^{[2]}, c_2^{[2]} > 0, n_2 > 0, \forall n \geq n_2, 0 \leq c_1^{[2]} \cdot h(n) \leq g(n) \leq c_2^{[2]} \cdot h(n)$$

Let $c_1^{[1]} = c_2^{[1]} = 1, f(n) = g(n)$,
$0 \leq c_1^{[2]} \cdot h(n) \leq f(n) \leq c_2^{[2]} \cdot h(n) \implies f(n) \in \Theta(h(n))$

5.4 $f(n) \in o(g(n)) \wedge g(n) \in o(h(n)) \implies f(n) \in o(h(n))$

Proof.

$$\forall c_1 > 0, \exists n_1 > 0, \forall n \geq n_1, 0 \leq f(n) < c_1 \cdot g(n) \tag{1}$$
$$\forall c_2 > 0, \exists n_2 > 0, \forall n \geq n_2, 0 \leq g(n) < c_2 \cdot h(n) \tag{2}$$

From (2), $\forall c_2 > 0, \exists n_2 > 0, \forall n \geq n_2, 0 \leq c_1 \cdot g(n) < c_1 \cdot c_2 \cdot h(n)$

$$\forall c_1, c_2 > 0, \exists n_1, n_2 > 0, \forall n \geq \max(n_1, n_2), 0 \leq f(n) < c_1 \cdot g(n) < c_1 \cdot c_2 \cdot h(n)$$
$$\forall c_1, c_2 > 0, \exists n_1, n_2 > 0, \forall n \geq \max(n_1, n_2), 0 \leq f(n) < c_1 \cdot c_2 \cdot h(n)$$

□

5.5 $f(n) \in \omega(g(n) \wedge g(n) \in \omega(h(n))) \implies f(n) \in \omega(h(n))$

$$\forall c_1 > 0, \exists n_1 > 0, \forall n \geq n_1, 0 \leq c_1 \cdot g(n) < f(n)$$
$$\forall c_2 > 0, \exists n_2 > 0, \forall n \geq n_2, 0 \leq c_2 \cdot h(n) < g(n)$$
$$\forall c_1, c_2 > 0, \exists n_1, n_2 > 0, \geq \max(n_1, n_2), 0 \leq c_1 \cdot c_2 \cdot h(n) < c_1 \cdot g(n) < f(n)$$

6 Symmetry: $f(n) \in \Theta(g(n)) \iff g(n) \in \Theta(f(n))$

Proof.

$$f(n) \in \Theta(g(n)) \implies$$
$$\exists c_1, c_2 > 0, n_1 > 0, \forall n \geq n_1, 0 \leq c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n) \implies$$
$$\left(0 \leq g(n) \leq \frac{1}{c_1} \cdot f(n) \leq \frac{c_2}{c_1} \cdot g(n)\right) \wedge \left(0 \leq \frac{c_1}{c_2} \cdot g(n) \leq \frac{1}{c_2} \cdot f(n) \leq g(n)\right) \implies$$
$$0 \leq \frac{1}{c_2} \cdot f(n) \leq g(n) \leq \frac{1}{c_1} \cdot f(n)$$

Identical proof for the other direction $\square$

7 Complementary

7.1 $f(n) \in O(g(n) \iff g(n) \in \Omega(f(n)))$

Proof.

$$\begin{aligned}
&f(n) \in O(g(n)) \implies \\
&\exists c > 0 \wedge n_0 > 0, \forall n \geq n_0, 0 \leq f(n) \leq c \cdot g(n) \implies \\
&0 \leq \frac{1}{c} \cdot f(n) \leq g(n)
\end{aligned}$$

Proof for the other direction and the next statement are identical. $\square$

7.2 $f(n) \in o(g(n) \iff g(n) \in \omega(f(n)))$

8 Problems

$3^{n+1} \in O(3^n)$

Proof.

$$\begin{aligned}
&3^{n+1} = 3 \cdot 3^n \\
&letc = 4, n_0 = 1, \forall n \geq n_0, \\
&3^{n+1} < c \cdot 3^n
\end{aligned}$$

$\square$

$4^n \notin O(2^n)$

Proof. Suppose $f(n) = 4^n \in O(2^n)$,

$$\begin{aligned}
&\exists c > 0, n_0 > 0, \forall n \geq n_0, 0 \leq 2^{2 \times n} \leq c \cdot 2^n \implies \\
&0 \leq 2^n \leq c \\
&0 \leq n \leq \log_2 c(const) \\
&\text{n is upper bounded by a constant, a contradiction}
\end{aligned}$$

$\square$

$2^{\lfloor \log_2 n \rfloor} \in \Theta(n)$

Proof.

$$\begin{aligned}
&\text{Let } c_1 = \frac{1}{2}, c_2 = 1, n_0 = 2, \forall n \geq n_0, \\
&0 \leq \frac{1}{2} \cdot 2^{\lfloor \log_2 n \rfloor} \leq \frac{1}{2} \cdot 2^{\log_2 n} \leq 2^{\lfloor \log_2 n \rfloor} \leq n
\end{aligned}$$

$\square$